Thermodynamic description of the Mg–Sn system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Mg–Sn binary system as calculated with pycalphad from the SGTE dataset MgSn-04Fri.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Mg) + Mg2Sn, lies at 563.4 °C.

1. Introduction

The Mg–Sn system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as congruent compound. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of S.G. Fries, unpublished work, 2004. [2]. The calculations use pycalphad 0.11.2 at 1 atm.

Pure Mg melts at 649.9 °C. For Sn, the liquidus meets the axis at 231.9 °C.

2. Phase diagram and invariant reactions

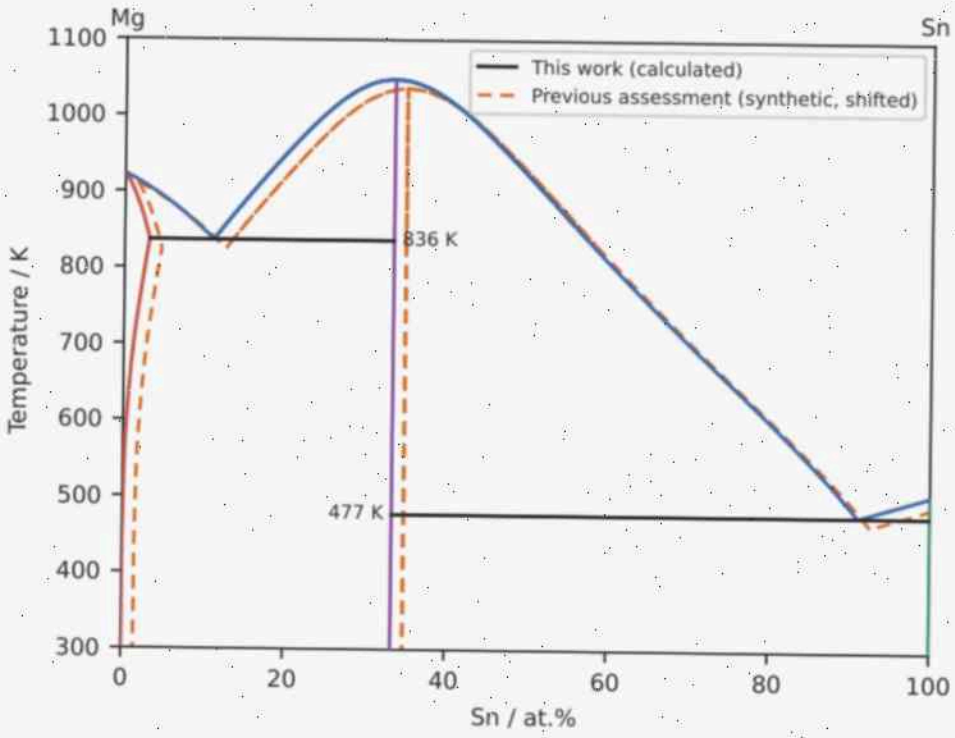


Fig. 1. Calculated Mg–Sn phase diagram and a dashed, deliberately shifted second assessment.

At 563.4 °C the eutectic L → (Mg) + Mg2Sn takes place; the L phase then contains x(Sn) = 0.110. At 204.0 °C the eutectic L → Mg2Sn + (Sn) takes place; the L phase then contains a mole fraction of 0.911. Mg2Sn melts congruently at 775.5 °C and x(Sn) = 0.333.

Table 1. Invariant reactions and congruent points, calculated (Mg–Sn).

Reaction	Type	T / K	Compositions (at.% Sn)		
		this work	phase 1	phase 2	phase 3
L → Mg2Sn	congruent melting[a]	1048.7	33.3	33.3	–
L → (Mg) + Mg2Sn	eutectic	836.5	11.0	3.1	33.3
L → Mg2Sn + (Sn)	eutectic	477.1	91.1	33.3	100.0

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.
[a] Two-phase equilibrium at a temperature extremum; both phases have the same composition.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Mg–Sn).

Phase	Structure			Stability range / K		Homogeneity range	
	Pearson	Strukturb.	Prototype	from	to	at.% Sn	at T / K
(Mg)	hP2	A3	Mg	< 300	923.0	0.0–3.0	840
Mg2Sn	cF12	C1	CaF2	< 300	1048.7	33.3	440
(Sn)	tI4	A5	β-Sn	< 300	505.1	100.0	300

Structure labels follow the TDB phase names (– where the name does not identify a structure). Stability range: temperatures between which the phase is stable in some part of the diagram.

The upper stability limit of Mg2Sn is 775.5 °C. The solubility of Sn in (Mg) reaches a mole fraction of 0.031 at 563.4 °C.

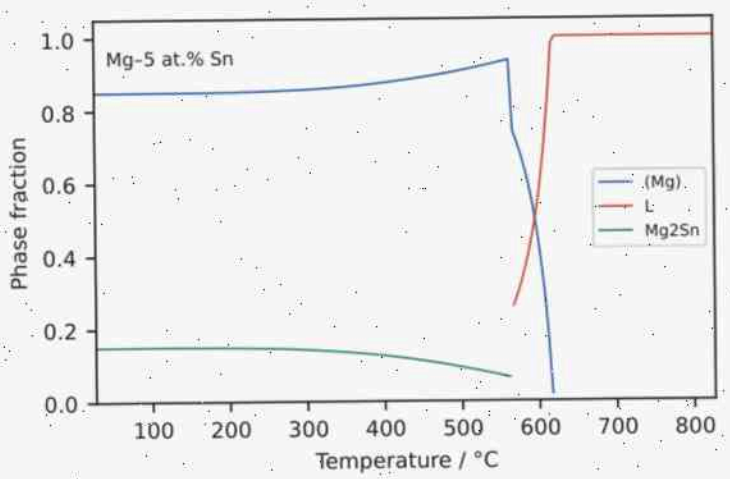


Fig. 2. Calculated equilibrium phase fractions versus temperature for Mg–5 at.% Sn.

4. Thermodynamic properties

At 1150 K the integral mixing enthalpy of the liquid has its minimum, −14298 J/mol, at x(Sn) = 0.380. The calculated enthalpy of formation of Mg2Sn at 298.15 K is −25.64 kJ/mol per mole of atoms. Its heat capacity at room temperature is 28.43 J/(mol K). For the equiatomic liquid at 1150 K, the Gibbs energy of mixing is −18.82 kJ/mol-atom.

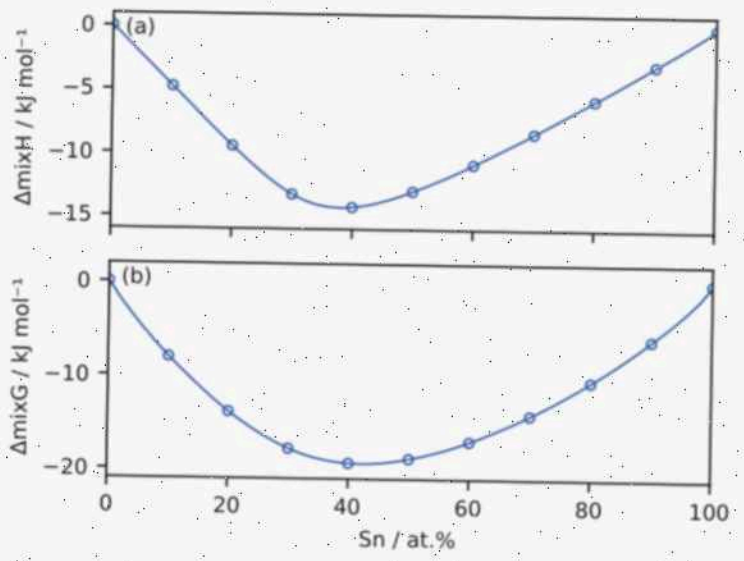


Fig. 3. Calculated (a) mixing enthalpy and (b) mixing Gibbs energy of the liquid at 1150 K.

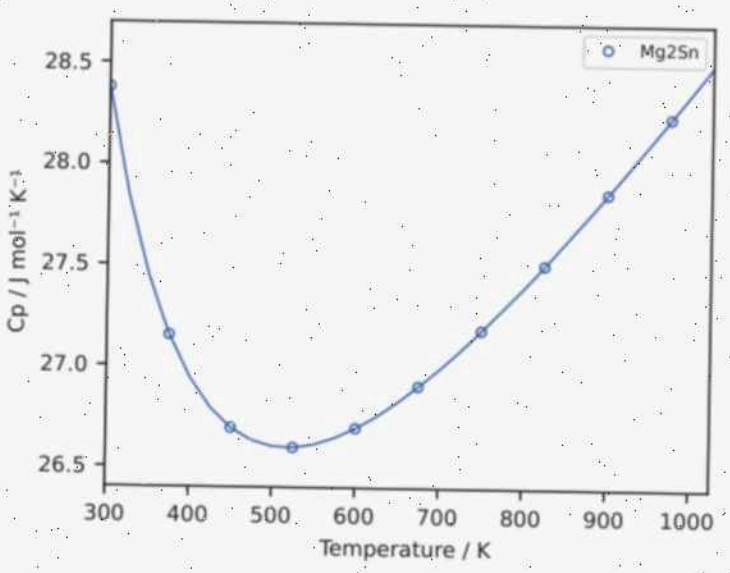


Fig. 4. Calculated heat capacity of Mg2Sn.

Table 3. Calculated formation properties of the intermediate phases at 298.15 K (Mg–Sn).

Phase	at.% Sn	ΔfH(298.15 K) / kJ mol⁻¹ (f.u.)	ΔfS / J mol⁻¹ K⁻¹ (f.u.)	Cp(298.15 K) / J mol⁻¹ K⁻¹ (f.u.)
Mg2Sn	33.3	−76.91 ± 0.5	−11.01	85.28

Reference: stable elements (SER) at 298.15 K; values per mole of formula units (formula unit as written in the Phase column). ± values are synthetic illustration uncertainties, not data.

Table 4. Integral and partial mixing properties of the liquid at 1150 K (Mg–Sn).

	Liquid at 1150 K			
at.% Sn	ΔmixH / kJ mol⁻¹	ΔmixG / kJ mol⁻¹	a(Mg)	a(Sn)
10	−4.76	−8.01	0.8722	7.866E−4
30	−13.25	−17.78	0.3711	0.0205
50	−12.98	−18.82	0.0717	0.2722
70	−8.36	−14.13	0.0213	0.6296
90	−2.95	−6.05	0.0050	0.8925

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] S.G. Fries, unpublished work, 2004.

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.